

- 5 The pressure of water vapour inside and outside leaves can be measured. The difference between these pressures is known as the leaf vapour pressure deficit (LVPD).

LVPD is one of the factors that influences the rate of transpiration.

Scientists measured the effect of changing the LVPD on the rate of transpiration in several species of flowering plant that live in a variety of different habitats. Two of these species were:

- *Nerium oleander*, a species that is adapted to grow in hot, dry conditions
- *Helianthus annuus*, a species that is not adapted for survival in hot, dry conditions.

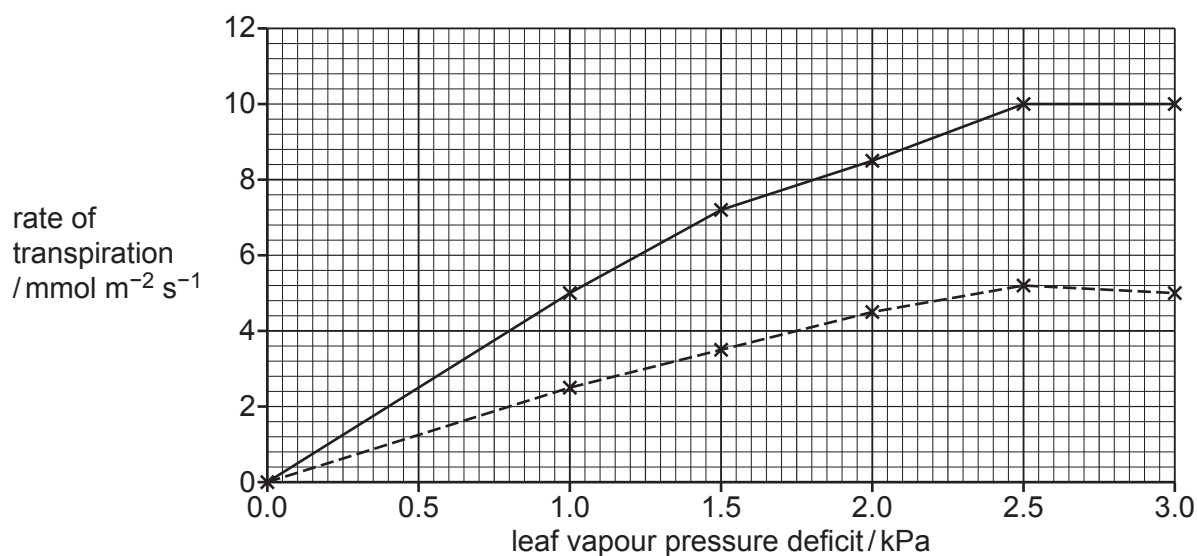
Fig. 5.1 shows the effect of increasing the LVPD on the transpiration rates of the two species.

All other factors were kept constant.

**key**

———— *Helianthus annuus*

----- *Nerium oleander*



**Fig. 5.1**

**(a)** Compare the results of the two species shown in Fig. 5.1.

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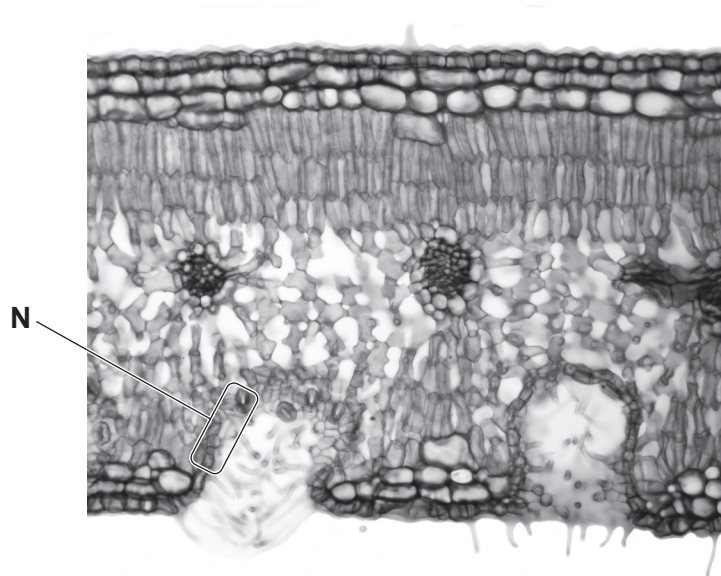
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(b) Fig. 5.2 shows part of a plant of *N. oleander*.



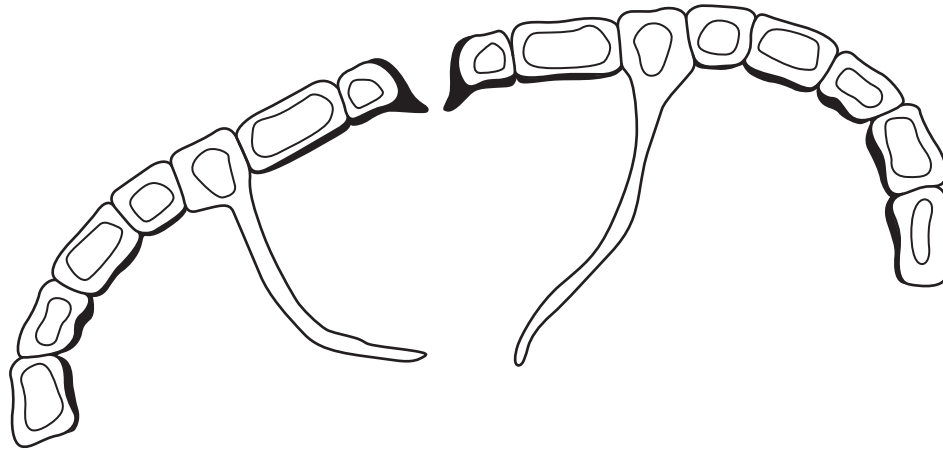
**Fig. 5.2**

Fig. 5.3 shows a cross-section of part of an oleander leaf.



**Fig. 5.3**

Fig. 5.4 is a drawing of a high-power view of region **N** on Fig. 5.3.



**Fig. 5.4**

State **and** explain **two** adaptations shown by the leaves of *N. oleander* that are visible in Fig. 5.3 and Fig. 5.4.

one adaptation visible in Fig. 5.3 .....

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explanation .....

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one adaptation visible in Fig. 5.4 .....

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explanation .....

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